Homework 4

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1 Goal

This homework proves that graphs typed over a fixed type graph TG, together with type-preserving graph morphisms, form a category. The proof first shows that graph morphisms are closed under composition. It then verifies composition, identities, associativity, and the identity laws for typed graphs.

2 Basic definitions

Definition 1 (Directed graph)**.** A directed graph is a tuple

$$G = (V_G, E_G, s_G, t_G),$$

where V_G is the set of vertices, E_G is the set of edges, and $s_G, t_G : E_G \to V_G$ give the source and target of every edge.

Definition 2 (Graph morphism)**.** Let G and H be directed graphs. A graph morphism $f : G \to H$ is a pair of functions

$$f_V : V_G \to V_H, \qquad f_E : E_G \to E_H$$

that preserve the source and target of every edge:

$$f_V \circ s_G = s_H \circ f_E, \qquad f_V \circ t_G = t_H \circ f_E. \tag{1}$$

The two equations say that mapping an edge and then reading its endpoint gives the same result as reading the original endpoint and then mapping that vertex. Therefore, a graph morphism preserves the incidence structure of the graph.

3 Composition of graph morphisms

The following lemma proves the explicit homework statement appearing earlier in the lecture material.

Lemma 1. *The composition of two graph morphisms is a graph morphism.*

Proof. Let

$$f = (f_V, f_E) : G \to H \qquad \text{and} \qquad g = (g_V, g_E) : H \to K$$

be graph morphisms. Define their composition componentwise:

$$g \circ f = (g_V \circ f_V,\ g_E \circ f_E) : G \to K.$$

We first verify preservation of sources. Using associativity of function composition and the morphism equations for f and g, we obtain

$$\begin{aligned}(g_V \circ f_V) \circ s_G &= g_V \circ (f_V \circ s_G)\\ &= g_V \circ (s_H \circ f_E)\\ &= (g_V \circ s_H) \circ f_E\\ &= (s_K \circ g_E) \circ f_E\\ &= s_K \circ (g_E \circ f_E).\end{aligned}$$

The same calculation for targets gives

$$(g_V \circ f_V) \circ t_G = t_K \circ (g_E \circ f_E).$$

Hence $g \circ f$ satisfies both equations in Equation 1, so it is a graph morphism. ∎

4 Typed graphs

Fix one graph TG, called the *type graph*.

Definition 3 (Typed graph)**.** A graph typed over TG is a pair $(G, type_G)$, where G is a graph and

$$type_G : G \to TG$$

is a graph morphism. It assigns a type in TG to every vertex and edge of G while preserving sources and targets.

Definition 4 (Type-preserving morphism)**.** A morphism between two graphs typed over the same TG,

$$f : (G, type_G) \to (H, type_H),$$

is a graph morphism $f : G \to H$ satisfying

$$type_H \circ f = type_G. \tag{2}$$

This equality means that the following triangle commutes:

$$\begin{tikzcd} G \arrow[rr, "f"] \arrow[dr, "type_G"'] & & H \arrow[dl, "type_H"] \\ & TG & \end{tikzcd}$$

In simple terms, f may map vertices and edges to different elements, but it cannot change their types.

5 The category of graphs typed over TG

We define a structure denoted by **Graph**$/TG$:

- its **objects** are all pairs $(G, type_G)$ with $type_G : G \to TG$;
- its **morphisms** are graph morphisms that satisfy Equation 2;
- composition is the componentwise composition of the underlying graph morphisms;
- the identity on $(G, type_G)$ is the ordinary graph identity id_G.

Theorem 1. *The structure* **Graph**$/TG$ *is a category.*

Proof. We verify the required properties.

Closure under composition. Consider two composable type-preserving morphisms

$$f : (G, type_G) \to (H, type_H), \qquad g : (H, type_H) \to (K, type_K).$$

By the lemma, the underlying composition $g \circ f : G \to K$ is a graph morphism. It also preserves typing because

$$\begin{aligned} type_K \circ (g \circ f) &= (type_K \circ g) \circ f && \text{by associativity} \\ &= type_H \circ f && \text{because } g \text{ preserves types} \\ &= type_G && \text{because } f \text{ preserves types.} \end{aligned}$$

Therefore, $g \circ f$ is a morphism in **Graph**$/TG$.

The corresponding commuting diagram is

$$\begin{tikzcd} G \arrow[r, "f"] \arrow[rr, bend left, "g \circ f"] \arrow[dr, "type_G"'] & H \arrow[r, "g"] \arrow[d, "type_H"] & K \arrow[dl, "type_K"] \\ & TG & \end{tikzcd}$$

Identity morphisms. For every object $(G, type_G)$, the graph identity $id_G : G \to G$ preserves typing:

$$type_G \circ id_G = type_G.$$

Thus, every object has a valid identity morphism in $\mathbf{Graph}/TG$.

Associativity. Typed morphisms are composed through their vertex and edge functions. Function composition is associative, so for all composable typed morphisms f, g, and h,

$$(h \circ g) \circ f = h \circ (g \circ f).$$

Closure, proved above, ensures that both sides are still type-preserving morphisms.

Identity laws. For every typed morphism $f : (G, type_G) \to (H, type_H)$, the identity laws of the underlying graph morphisms give

$$id_H \circ f = f \qquad \text{and} \qquad f \circ id_G = f.$$

The identities and f preserve typing, so these equalities hold inside $\mathbf{Graph}/TG$.

All the category requirements are satisfied. Therefore, graphs typed over the fixed graph TG, with type-preserving graph morphisms, form a category. □

6 Categorical interpretation

The category just constructed is the *slice category* of graphs over TG. An object is an arrow $G \to TG$, and a morphism is an arrow between their domains that makes the typing triangle commute. This is commonly written as $\mathbf{Graph}/TG$ and is also represented in the lecture slides as $\mathbf{Graph} \downarrow TG$.

This interpretation explains the construction at a more general level: for any category $\mathbf{C}$ and fixed object X, the arrows with codomain X and their commuting triangles form the slice category $\mathbf{C}/X$.

7 Conclusion

The essential point is that all operations remain compatible with the fixed typing graph. Ordinary graph morphisms compose, and the equation $type_H \circ f = type_G$ remains true after composition. Identity morphisms also preserve typing, while associativity and the identity laws are inherited componentwise from ordinary functions. Consequently, $\mathbf{Graph}/TG$ is a well-defined category.